Homework 4: Circle flows

MATH 515 — Fall 2026

Due Friday, October 23, 2026

Assignment convention. Each numbered problem reproduces the full assigned statement from Steven H. Strogatz, *Nonlinear Dynamics and Chaos*, third edition (the specific section/problem number is given parenthetically), with some mild editing for clarity. Any additional requirements are labeled **Course-specific instructions** and are part of the assignment.

Problem 1: Potentials on the circle

Textbook problem (Strogatz, third edition, Exercise 4.1.8, "Potentials for vector fields on the circle"):

(a) Consider the vector field on the circle given by $\dot{\theta} = \cos\theta$. Show that this system has a single-valued potential $V(\theta)$, i.e., for each point on the circle, there is a well-defined value of V such that $\dot{\theta} = -\,\mathrm{d}V/\,\mathrm{d}\theta$. (As usual, θ and $\theta + 2\pi k$ are to be regarded as the same point on the circle, for each integer k.)

(b) Now consider $\dot{\theta} = 1$. Show that there is no single-valued potential $V(\theta)$ for this vector field on the circle.

(c) What's the general rule? When does $\dot{\theta} = f(\theta)$ have a single-valued potential?

Course-specific instructions. Complete all parts. State the general criterion in a form involving one full-period integral, and prove both directions. In part (c), assume that $f : \mathbb{R} \to \mathbb{R}$ is continuous and 2π-periodic.

Problem 2: Several saddle-node bifurcations on a circle [Challenge]

Textbook problem (Strogatz, third edition, Exercise 4.3.5):

Draw the phase portrait as a function of the control parameter μ. Classify the bifurcations that occur as μ varies, and find all the bifurcation values of μ, for

$$\dot{\theta} = \mu + \cos\theta + \cos 2\theta.$$

Course-specific instructions. Complete the exercise for all real μ. A labeled circle sketch for each parameter regime is sufficient; give the equilibrium angles in terms of arccos rather than decimal approximations. If two saddle-node bifurcations occur at the same parameter value, identify their locations and explain why they occur together.

Problem 3: A circle flow with a triangle-wave response

Textbook problem (Strogatz, third edition, Exercise 4.5.1, "Triangle wave (adapted)"):

Consider the circle flow

$$\dot{\phi} = \Delta - Af(\phi),$$

where f is the triangle wave defined by

$$f(\phi) = \begin{cases} \phi, & -\dfrac{\pi}{2} \le \phi \le \dfrac{\pi}{2}, \\ \pi - \phi, & \dfrac{\pi}{2} \le \phi \le \dfrac{3\pi}{2} \end{cases}$$

on the interval $-\pi/2 \le \phi \le 3\pi/2$, and extend f periodically outside this interval.

(a) Graph $f(\phi)$.

(b) Find the range of Δ for which equilibria exist.

(c) Find all equilibrium phases ϕ^* modulo 2π and determine their stability, including at the endpoints of the range in part (b).

(d) When no equilibria exist, find a formula for T_{drift}, defined to be the positive elapsed time for ϕ to make one complete revolution around the circle.

Course-specific instructions. Assume $A > 0$ and $\Delta \in \mathbb{R}$. Complete all parts. In part (d), distinguish the cases $\Delta > 0$ and $\Delta < 0$.

Problem 4: Excitability on a circle

Textbook problem (Strogatz, third edition, Exercise 4.5.3, "Excitable systems"):

Suppose you stimulate a neuron by injecting it with a pulse of current. If the stimulus is small, nothing dramatic happens: the neuron increases its membrane potential slightly, and then relaxes back to its resting potential. However, if the stimulus exceeds a certain threshold, the neuron will "fire" and produce a large voltage spike before returning to rest. Surprisingly, the size of the spike doesn't depend much on the size of the stimulus—anything above threshold will elicit essentially the same response.

Similar phenomena are found in other types of cells and even in some chemical reactions (Winfree 1980, Rinzel and Ermentrout 1989, Murray 2002). These systems are called *excitable*. The term is hard to define precisely, but roughly speaking, an excitable system is characterized by two properties: (1) it has a unique, globally attracting rest state, and (2) a large enough stimulus can send the system on a long excursion through phase space before it returns to the resting state.

This exercise deals with the simplest caricature of an excitable system. Let $\dot{\theta} = \mu + \sin\theta$, where μ is slightly less than 1.

(a) Show that the system satisfies the two properties mentioned above. What object plays the role of the "rest state"? And the "threshold"?

(b) Let $V(t) = \cos\theta(t)$. Sketch $V(t)$ for various initial conditions. (Here V is analogous to the neuron's membrane potential, and the initial conditions correspond to different perturbations from the rest state.)

Course-specific instructions. Throughout, take $0 < \mu < 1$ close to 1.

For part (a), the textbook's claim of global attraction is slightly incorrect: there is indeed an initial phase that does not approach the rest state equilibrium. However, it is *very nearly* globally attracting (you will see what I mean by this when you work through the problem). Complete part (a) as instructed with this small caveat in mind.

For part (b), draw the phase portrait on the circle, and let θ_s denote the stable equilibrium found in part (a). Model a stimulus of size $q > 0$ as an instantaneous increase in phase from θ_s to $\theta_s + q$, so that immediately after the stimulus is applied,

$$\theta(0) = \theta_s + q \pmod{2\pi}.$$

For $t > 0$, the phase then is just the usual solution to the differential equation. Find the threshold stimulus size q_{crit}: the smallest positive value of q that places the system exactly at the unstable equilibrium. Use the phase portrait to compare what happens when $0 < q < q_{\text{crit}}$ and when q is just above q_{crit}, and sketch a graph of $V(t) = \cos\theta(t)$ in part (b). Also explain what happens when $q = q_{\text{crit}}$.